VII. Expectations, Moments, and Moment Generating Functions (GSE 518)

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Idea: Expectations play a key role in statistics/econometrics.

1. Expectations

Definition: The expectation of r.v. X is

$$E(X) = \begin{cases} \sum_{x \in \mathcal{X}} x f_X(x) & \text{where } f \text{ is a pmf if discrete} \\ \int_{x \in \mathcal{X}} x f_X(x)\, dx & \text{where } f \text{ is a pdf if continuous} \end{cases}$$

provided the sum/integral exists or else $E(X)$ will not exist.

Interpretation of $E(X)$:

- It's a weighted average of X with weight that is equal to $f_X(x)$

- Can think of $E(X)$ as a "long run average."

Say if we take n random draws of X from $f_X(x)$, then (informally)

$$\text{“} \lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} x_i = E(X) \text{”}$$

can be thought of as the expected average.

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

- It is a measure of the *center* of the distribution.

Example (Discrete Case): Let $X \sim f_X(x)$ with

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

for $1 > p > 0$.

Find $E(X)$.

$$E(X) = \sum_{x=0}^{1} x f_X(x) = 0 f_X(0) + 1 f_X(1) = 0(1-p) + 1(p) = p$$

Continuing the example:

Flip a coin, let $X = 1$ if heads, then X follows a Bernoulli distribution with $p = \frac{1}{2}$.

$$E(X) = \frac{1}{2}$$

If you flip the coin n times, we expect to see the ratio of heads to total flips to approach $\frac{1}{2}$.

Example (Continuous Case): Let $X \sim \text{uniform}(0, 1)$,

$$f_X(x) = \begin{cases} 1 & \text{if } x \in (0,1) \\ 0 & \text{otherwise} \end{cases}$$

Find $E(X)$.

$$E(X) = \int_0^1 x f_X(x)\, dx = \int_0^1 x(1)\, dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

1.1. General Expectations

The expected value of $g(x)$ is:

$$E(g(X)) = \begin{cases} \sum\limits_{x \in \mathcal{X}} g(x) f_X(x) & \text{if } X \text{ is discrete} \\ \int\limits_{x \in \mathcal{X}} g(x) f_X(x)\, dx & \text{if } X \text{ is continuous} \end{cases}$$

if sum/integral exists.

Example:

$$E(X^2) = \begin{cases} \int\limits_{x \in \mathcal{X}} x^2 f_X(x)\, dx & \text{if continuous} \\ \sum\limits_{x \in \mathcal{X}} x^2 f_X(x) & \text{if discrete} \end{cases}$$

2. Properties of Expectations

In general:

$$E(g(X)) \neq g(E(X))$$

Example:

$$E(X^2) \neq [E(X)]^2 \quad \text{or} \quad E(e^X) \neq e^{E(X)}$$

Analogy: $X^2 + Y^2 \neq (X + Y)^2$

Or

$$\int\limits_{x \in \mathcal{X}} x^2 f_X(x)\, dx \neq \left(\int\limits_{x \in \mathcal{X}} x f_X(x)\, dx \right)^2$$

Key Property of $E(x)$: Linearity

Let a and b be constants.

$$E(aX + b) = aE(X) + b$$

Note:

- Can take a constant outside expected value $E(aX) = aE(X)$
- $E(c) = c$ if c is a constant.

3. Higher Moments

Definition:

For each integer n, the *n-th moment* of X is:

$$M_n' = E(X^n)$$

And the *n-th central moment* is:

$$M_n = E\left[(X - \mu)^n\right],$$

where: $\mu = M_1 = M_1' = E(X)$

Key moments:

$$E(X), \quad E\left[(X - \mu)^2\right]$$

3.1. Variance

Definition of Variance:

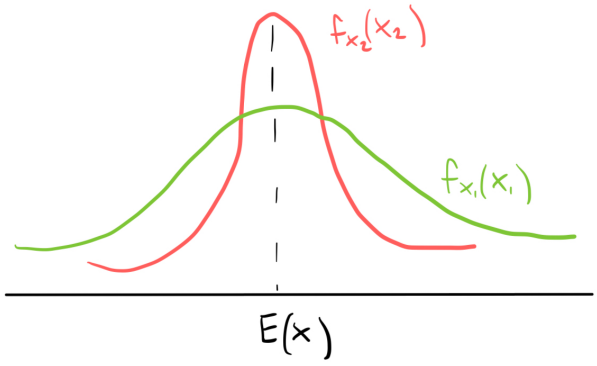
Let $\mu = E(X)$. The variance of a random variable X is

$$\mathrm{Var}(X) = E\left[(X - \mu)^2\right],$$

and its standard deviation of X is $\sqrt{\mathrm{Var}(X)}$.

Notes:

1. The variance measures the dispersion of X about its mean $E(X)$.



   $f_{X_1}(x_1)$ has high variance (high chance to get a value x away from $E(X)$).

   $f_{X_2}(x_2)$ has low variance (low chance to get a value x away from $E(X)$).

2. $\sqrt{\text{Var}(X)}$ has same units as X but $\text{Var}(X)$ is in (units)^2.

3. For constants a and b:

$$\text{Var}(aX+b) = a^2\text{Var}(X)$$

   Proof:

$$\text{Var}(aX+b) = E\left[(aX+b-E(aX+b))^2\right] \text{ (by definition of variance)}$$

$$= E\left[(aX+b-aE(X)-b)^2\right] \text{ (apply the linearity property of expectations)}$$

$$= E\left[[a(X-E(X))]^2\right] = E\left[a^2(X-E(X))^2\right]$$

$$= a^2E\left[(X-E(X))^2\right] = a^2\text{Var}(X)$$

4. $\text{Var}(X) = E(X^2) - [E(X)]^2$

   Proof:

$$\text{Var}(X) = E\left[(X - E(X))^2\right]$$

$$= E\left[X^2 - 2XE(X) + [E(X)]^2\right]$$

   (*Key Note*: It is always critical to keep track of what is a *random variable* and what is a *constant*. X is random variable but $E(X)$ is constant!)

$$= E(X^2) - 2E(X)E(X) + [E(X)]^2$$

$$= E(X^2) - [E(X)]^2$$

5. The variance of $g(X)$, a general function of X, is given by:

$$Var(g(X)) = E\left[\{g(X) - E(g(X))\}^2\right] = E\left[\{g(X)\}^2\right] - [E\left(g(X)\right)]^2$$

Example 1:

Let $X \sim \text{Bernoulli}(p)$ distribution. Find $\text{Var}(X)$.

From before, we know the pmf is given by:

$$f_x(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

and $E(x) = p$ (we found $E(x)$ before).

To find $Var(X)$, we first get $E(X^2)$ so we can use the formula $\text{Var}(X) = E(X^2) - [E(X)]^2$:

$$E(X^2) = \sum_{x=0}^{1} x^2 f_x(x) = 0^2 f_x(0) + 1^2 f_x(1) = p$$

so

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1-p)$$

Example 2:

Flip a coin. Let $X = 1$ if heads. We know $p = \frac{1}{2}$. Above we found that

$$E(X) = \frac{1}{2} \quad \text{and} \quad \text{Var}(X) = p(1-p)$$

Thus,

$$\text{Var}(x) = \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$\text{S.D.} = \sqrt{\text{Var}(x)} = \frac{1}{2}$$

4. Moment Generating Function (mgf)

Definition:

Let $X \sim f_X(x)$. The *moment generating function mgf* of the random variable X, $M_X(t)$, is given by:

$$M_X(t) = E\left[e^{tX}\right],$$

provided that sum/integral exists in a neighborhood around $t = 0$.

Explicitly:

$$M_X(t) = E\left[e^{Xt}\right] = \begin{cases} \sum\limits_{x\in\mathcal{X}} e^{tx} f_X(x), & \text{if } X \text{ is discrete} \\ \int\limits_{x\in\mathcal{X}} e^{tx} f_X(x)\, dx & \text{if } X \text{ is continuous} \end{cases}$$

Example (Discrete Case):

Same as before: Let $X \sim \text{Bernoulli}(p)$

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the mgf:

$$M_X(t) = E\left[e^{tX}\right] = \sum_{x=0}^{1} e^{tx} f_X(x) = e^{t0} f_X(0) + e^{1t} f_X(1)$$

$$= (1-p) + e^t p$$

$$\boxed{M_X(t) = (1-p) + e^t p}$$

Example (Continuous Case):

Get mgf X when of $X \sim \text{Exponential}(\lambda)$, $\lambda > 0$ with pdf

$$f_X(x) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \int_0^\infty e^{tx} f_X(x)\,dx = \int_0^\infty e^{tx}\frac{1}{\lambda}e^{-\frac{x}{\lambda}}\,dx$$

$$= \int_0^\infty \frac{1}{\lambda}e^{-\frac{x(1-t\lambda)}{\lambda}}\,dx = -\frac{1}{1-t\lambda}e^{-\frac{x(1-t\lambda)}{\lambda}}\Big|_0^\infty = \frac{1}{1-t\lambda}, \quad \text{provided } 1-t\lambda > 0$$

$$\boxed{M_X(t) = \frac{1}{1-t\lambda} \quad \text{provided } 1-t\lambda > 0, \text{ or } \frac{1}{\lambda} > t}$$

(Note: For the continuous example, do not worry about how to explicitly solve that particular integral, say, the second line in the derivation above. Please do understand the first line, i.e., how to go from the definition of the mgf (i.e., $M_X(t)$) to the explicit intergral we would need to solve to get it (i.e., $\int_0^\infty e^{tx}\frac{1}{\lambda}e^{-\frac{x}{\lambda}}\,dx$).

3. Uses of the mgf

There are three main uses of the mgf:

a) To generate moments (get)

b) To characterize the *exact* distribution of X (like the cdf/pmf/pdf)

c) To obtain an *approximation* of the distribution of X

a) To get moments

Theorem: (*Theorem 2.3.7. in textbook*) If X has *mgf* $M_X(t)$, then:

$$E\left(X^n\right) = M_X^{(n)}(0) = \left.\frac{d^n M_X(t)}{dt^n}\right|_{t=0}$$

Example 1: Use *mgf* of Bernoulli(p) to get $E(X)$ and $E(X^2)$

From our above example: $M_X(t) = (1-p) + pe^t$. Then,

$$E(X) = \left.\frac{dM_X(t)}{dt}\right|_{t=0} = \left.pe^t\right|_{t=0} = p$$

$$E(X^2) = \left.\frac{d^2M_X(t)}{dt^2}\right|_{t=0} = \left.pe^t\right|_{t=0} = p$$

Example 2: Find $\text{Var}(X)$ when $X \sim \text{Exponential}(\lambda)$.

From our above example: $M_X(t) = (1-\lambda t)^{-1}$. Also, we know that

$\text{Var}(X) = E(X^2) - [E(X)]^2$. Then,

$$E(X) = \left.\frac{dM_X(t)}{dt}\right|_{t=0} = \left.(-1)(1-\lambda t)^{-2}(-\lambda)\right|_{t=0} = \lambda$$

$$E(X^2) = \left.\frac{d^2M_X(t)}{dt^2}\right|_{t=0} = \left.2(1-\lambda t)^{-3}\lambda^2\right|_{t=0} = 2\lambda^2$$

$$\text{Var}(X) = 2\lambda^2 - (\lambda)^2 = \lambda^2$$

b) To characterize the *exact* distribution of a random variable X

Theorem: (*Theorem 2.3.11. in textbook*) Suppose X has *mgf* $M_X(t)$ and Y has *mgf* $M_Y(t)$ and $M_X(t) = M_Y(t)$ for all t in a positive interval around 0. Then $X \sim Y$.

Informally: If two random variables have the same mgf, this implies they have the *same* distribution (i.e., they are identically distributed).

Example:

- Let $X \sim \text{Uniform}(0,1)$ with $f_X(x) = \begin{cases} 1 & \text{if } x \in (0,1) \\ 0 & \text{otherwise} \end{cases}$

- Also, let $Y = -\ln(X)$

- *Question:* Show that $Y \sim \text{Exponential}(1)$ with $\lambda = 1$ using mgf.

Remember from before that the mgf of an Exponential(λ) random variable is $M_X(t) = \frac{1}{1-\lambda t}$.

Plan: Find the mgf of Y and show that it equals the mgf of an Exponential(1) random variable.

$$M_Y(t) = E\left[e^{tY}\right] \text{ by def of mgf}$$

$$= E\left[e^{-\ln(X)t}\right] \text{ since } Y = -\ln(X)$$

$$= \int_0^1 e^{-\ln x \cdot t} \cdot (1)\, dx$$

(Recall that $a\ln(x) = \ln(x^a)$)

$$= \int_0^1 x^{-t}\, dx = \left[\frac{1}{-t+1}x^{-t+1}\right]\Bigg|_{x=0}^{x=1}$$

$$= \frac{1}{1-t} \text{ showing what we wanted.}$$

$$M_Y(t) = \frac{1}{1-t}$$

$$= \frac{1}{1-\lambda t} \text{ for exponential distribution.}$$

Since the mgf of Y equals the mgf of an exponential distribution with $\lambda = 1$,

$\Rightarrow Y \sim \text{Exponential}(1)$

c) To *approximate* distribution of a random variable X

Theorem: (*Theorem 2.3.12. in textbook*)

Suppose $\{X_n\}_{n=1}^{\infty}$, a sequence of r.v.s with mgfs $\{M_n(t)\}_{n=1}^{\infty}$, and $M_n(t)$ converge converges to $M_X(t)$ $\forall t$ around 0, where $M_X(t)$ is the mgf of a r.v. X.

Then

$$F_{X_n}(r) \to F_X(r) \quad \forall r \in \mathbb{R}.$$

Informally: Convergence of mgfs implies convergence of distributions.

Examples:

a) Poisson approximation to Binomial Distribution (college statistics)

b) Normal distribution approximation to Binomial Distribution (college statistics)

c) *Central Limit Theorem:* Very important result used to *approximate* the distribution of many estimators (later).

5. Useful Key Inequalities

1) Markov's Inequality

Let X be a non-negative random variable with finite mean. Then for any $c > 0$,

$$P(X \geq c) \leq \frac{E(X)}{c}$$

Proof: Assume X is a continuous r.v. (for discrete case replace integrals with sums):

$$E(X) = \int_0^\infty x \cdot f_X(x)\,dx = \int_0^c x \cdot f_X(x)\,dx + \int_c^\infty x \cdot f_X(x)\,dx \geq \int_c^\infty x \cdot f_X(x)\,dx$$

$$\geq \int_c^\infty c \cdot f_X(x)\,dx = c\int_c^\infty f_X(x)\,dx = c \cdot P(X \geq c)$$

Thus,

$$E(X) \geq c \cdot P(X \geq c) \quad \text{as desired.}$$

2) Chebyshev's Inequality

For any random variable Y with mean μ and variance σ^2, and any $k > 0$:

$$P(|Y - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Proof:

The proof here uses Markov's inequality:

$$P(X \geq c) \leq \frac{E(X)}{c}$$

To do so, let $X = (Y - \mu)^2$, which implies $E(X) = E[(Y - \mu)^2] = \text{Var}(Y)$. Also, let $c = k^2$. Applying Markov's inequality for that X and c we have:

$$P((Y - \mu)^2 \geq k^2) \leq \frac{\text{Var}(Y)}{k^2},$$

which is equivalent to

$$P(|Y - \mu| \geq k) \leq \frac{\text{Var}(Y)}{k^2} \quad \text{as desired.}$$

Uses of these inequalities:

- To bound probabilities
- Chebyshev's inequality is used to prove the *Law of Large Numbers* (later).